

EUGENE LEE

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Cincinnati, Ohio

SUMMARY

AI Researcher specializing in efficient deep learning, vision-language models, and in-context learning for LLM and VLM systems. Expertise in model compression, resource-constrained architectures, and multimodal alignment with focus on reducing inference costs and context window requirements. Proven track record of translating research into production-ready solutions with real-world impact.

Skills: PyTorch, **Python**, C/C++, CUDA, TypeScript/JavaScript, MATLAB, Verilog, AWS, Docker, iOS, Android, SQL, Redis

EDUCATION

National Yang Ming Chiao Tung University, Hsinchu *September 2017 - July 2023*

Ph.D. in Electronics Engineering, Advisor: Prof. Chen-Yi Lee; GPA: 4.0/4.0

National Chiao Tung University, Hsinchu *August 2013 - June 2017*

B.S. in Electronics Engineering

WORK EXPERIENCE

University of Cincinnati, Ohio *August 2024 – Present*

Postdoctoral Researcher, College of Medicine

- Developed novel **in-context learning** framework using Large Language Models to assess drug effects through **few-shot demonstrations**, achieving competitive performance while reducing labeled data requirements by 10x.
- Researched **demonstration selection strategies** that improve model reasoning quality and reduce context window size by 40%, directly addressing efficiency challenges in LLM inference.
- Built optimized imaging pipelines with temporal modeling for biological workflows, demonstrating expertise in **efficient multi-modal system design**.

[Paidge, Inc.](#)

January 2024 – Present

Founder

- Created an object-intelligence platform combining vision encoders with LLMs for conversational object interaction, deploying **efficient Vision-Language Models (VLMs)** using vLLM for production inference.
- Designed proprietary multi-frame aggregation model with FAISS for **large-scale instance retrieval**, optimizing embedding quality while maintaining sub-100ms query latency.
- Re-engineered lightweight GraphRAG for **high-throughput indexing**, reducing context requirements through structured knowledge representation and achieving 3x faster inference.
- Demonstrated expertise in translating research prototypes into production systems with measurable ROI and real-world deployment.

National Yang Ming Chiao Tung University, Hsinchu

August 2023 – July 2024

Postdoctoral Researcher, Institute of Electronics

- Directed architectural study optimizing **Vision-Language Pipelines** to improve token efficiency, reducing visual token count by 60% through **adaptive token reduction** while maintaining task performance.

- Researched **efficient attention mechanisms** for VLMs that minimize computational overhead in long-context scenarios, achieving 2x speedup in end-to-end latency.
- Led design of BioFPGA platform integrating CMOS ASIC with embedded systems, demonstrating hardware-software co-design expertise.

Advanced Bio Chips, Hsinchu
CEO

August 2023 – July 2024

- Co-led BioFPGA commercialization, securing approximately \$1M in funding and driving cross-functional integration across IC design, embedded software, and bioassay teams.
- Delivered system architecture optimizations that improved inference throughput by 5x while meeting strict power and memory constraints for edge deployment.

PROJECTS

In-Context Learning and Demonstration Optimization

August 2024 - Present

- Investigating **demonstration selection** strategies that maximize LLM reasoning performance while minimizing context window usage, achieving context reduction with improved task accuracy.
- Developed methods for **adaptive few-shot learning** that dynamically select optimal demonstrations based on query similarity, reducing inference costs.
- Exploring **implicit in-context learning** approaches that compress demonstration information into activation vectors, enabling zero-shot inference costs with few-shot performance.

Efficient Vision-Language Model Architectures

January 2023 - July 2024

- Designed token-efficient VLM architectures that reduce visual token count by 60% through **adaptive resolution** and **spatial pruning** techniques.
- Researched **multimodal alignment** methods that improve vision-language grounding while maintaining computational efficiency for real-time applications.
- Developed efficient cross-attention mechanisms for VLMs that scale to long-context scenarios with sub-linear computational complexity.

Foveated Video Super-Resolution with Hardware Acceleration

August 2021 - January 2022

- Applied complex VSR model with **spatially-adaptive computation** to meet bandwidth constraints in high-performance AR/VR systems.
- Integrated eye-tracking hardware with algorithm pipeline, conducting user studies to validate perceptual quality metrics and system performance.
- Led technical design and implementation of **FPGA hardware kernels** to accelerate computationally intensive tasks, achieving 10x speedup over software implementation.

Efficient Autonomous Learning Architectures

August 2018 - July 2024

- Developed **resource-constrained model architectures** that achieve competitive performance with 40% fewer parameters through neural architecture search and pruning.
- Pioneered **NeuralScale** method for efficient neuron scaling, enabling adaptive model capacity based on available computational resources (CVPR 2020 Oral).
- Researched **meta-learning** approaches for few-shot adaptation, demonstrating fast learning from limited demonstrations (ECCV 2020, ICCV 2021).

Knowledge Distillation and Model Compression

August 2018 - July 2023

- Developed **distillation techniques** for compressing large vision models into efficient student networks, retaining 95% of teacher performance with 3x fewer parameters.

- Researched **continual learning** methods that prevent catastrophic forgetting while maintaining model efficiency through modular architectures.
- Applied compression techniques to deploy models on resource-constrained wearable devices with strict power and memory budgets.

TEACHING EXPERIENCE

Introduction to VLSI Design

2019 Fall

Head Teaching Assistant

Instructor: Chen-Yi Lee

- This course aims to convey junior EE students techniques to analyze and design system by means of VLSI technology and CAD tools. Starting from VLSI process technology and transistor's behavior.

Integrated Circuit Design Laboratory

2018 Spring / 2017 Fall

Head Teaching Assistant / Teaching Assistant

Instructor: Chen-Yi Lee

- This course aims to convey the senior and graduated EE students techniques to design the VLSI chips using state-of-the-art CAD tools. In addition to learning CAD tools for performance-driven and cost-effective IC designs, a top-down design flow and related environment will also be addressed.

Introduction to Machine Learning

2018 Spring

Teaching Assistant

Instructor: Chen-Yi Lee

- This course introduces the fundamental concepts and algorithms that enable computers to learn from experience, with an emphasis on their practical application to real problems.

ACADEMIC ACHIEVEMENTS

Publications

h-index: 6; Citations: 423

Journal

- Chen, Rui, Eugene Lee, Yuxin Wang, Aditya Yadav, Minling Zhong, Pragti, Yujie Sun, and Jiajie Diao. **A Versatile Near-Infrared Fluorescent Probe for Fast Assessment of Lysosomal Status via a Large Multimodal Model.**, Aggregate: e70118, 2025.
- Yadav, Aditya, Eugene Lee, Rui Chen, Soryong R. Chae, Yujie Sun, and Jiajie Diao. **Understanding the Role of Polyethylene Glycol Coating in Reducing the Subcellular Toxicity of MXene Nanoparticles Using a Large Multimodal Model.**, Materials Today Bio (2025): 102372.
- Wang, Yuxin, Yun-Sheng Chan, Eugene Lee, Donglu Shi, Chen-Yi Lee, and Jiajie Diao, **Monitoring Escherichia Coli in Water through Real-Time Loop-Mediated Isothermal Amplification on Biochips**, Micromachines, vol. 15, no. 9, 2024, article 1112.
- Eugene Lee and Chen-Yi Lee, **PPG-Based Smart Wearable Device with Energy-Efficient Computing for Mobile Health-care Applications**, IEEE Sensors Journal, vol. 21, no. 12, pp. 13564-13573, 15 June, 2021, doi: 10.1109/JSEN.2021.3069460.

Conference

- Hsieh, Cheng-Hsuan, Eugene Lee, Jiajie Diao, Chen-Yi Lee, “**M*: On-Chip Microfluidic Operations with A* for Portable Diagnostics**,” In Proceedings of the International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025.
- Eugene Lee, Lien-Feng Hsu, Evan Chen and Chen-Yi Lee, “**Cross-Resolution Flow Propagation for Foveated Video Super-Resolution**,” In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), January 2023
- Eugene Lee, Cheng-Han Huang and Chen-Yi Lee, “**Few-Shot and Continual Learning with Attentive Independent Mechanisms**,” In Proceedings of the International Conference on Computer Vision (ICCV), October 2021

- Eugene Lee, Evan Chen and Chen-Yi Lee, “**Meta-rPPG: Remote Heart Rate Estimation Using a Transductive Meta-Learner**,” In Proceedings of the European Conference on Computer Vision (**ECCV**), August 2020
- Eugene Lee and Chen-Yi Lee, “**NeuralScale: Efficient Scaling of Neurons for Resource-Constrained Deep Neural Networks**,” In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), June 2020 (**Oral**)
- Eugene Lee, Annie Ho, Yi-Ting Wang, Cheng-Han Huang and Chen-Yi Lee, “**Cross-Domain Adaptation for Biometric Identification Using Photoplethysmogram**,” International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), May 2020
- Eugene Lee, Tsu-Jui Hsu and Chen-Yi Lee, “**Centralized State Sensing Using Sensor Array on Wearable Device**,” International Symposium on Circuits and Systems (**ISCAS**), May 2019
- Eugene Lee, Tsu-Jui Hsu and Chen-Yi Lee, “**Continuous Blood Glucose Monitoring on Wearable Device using Photoplethysmography**,” NSF IoT Workshop (**ICCAD**), Nov 2018

Awards

- Founder University Cohort 11 (LAUNCH Pre-Accelerator), 2025
- [Future Tech Awards, 2024](#)
- Draper University Fundamentals of Entrepreneurship 2022
- Novatek Ph.D. Fellowship, 2020~2022
- Recipient of Broadcom Foundation Scholarship, 2019
- 3rd place in Synopsys ARC Contest, 2017
- Award recipient of International ICCAD Contest, 2016

Academic Service

- Reviewer: CVPR [2022: 2; 2023: 3; 2025: 2], ICCV [2025: 4], ECCV [2022: 3; 2024: 1], NeurIPS [2025: 2], WACV [2026: 2], DeepMTL@ICCV [2021: 1], TCAS II [2020: 3].
- Talk: “*Semiconductor Chips for Fast Medical Testing*” at the 2024 Taiwan-Malaysia Semiconductor Forum, Kuala Lumpur, Malaysia.
- [Talk](#): “*Going Beyond Neural Architecture Search for Efficient Training and Inference*” at Academia Sinica, 2020.
- Talk: “*NeuralScale: Efficient Scaling of Neurons for Resource-Constrained Deep Neural Networks*” at National Taiwan University, 2020.

PATENTS

Lee, Chen-Yi, and Eugene Lee. **Video display systems, portable video display apparatuses and video enhancement methods**. U.S. Patent 12,211,174, issued January 28, 2025.

Lee, Chen-Yi, Eugene Lee, and Tsu Jui Hsu. **Physiological sensing method and device using the same**. U.S. Patent Application 16/441,801, filed July 23, 2020.